UNIT-V: Application Layer

Domain Name Space (DNS), DDNS, TELNET, EMAIL, File Transfer Protocol (FTP), WWW, HTTP, SNMP, Firewalls.

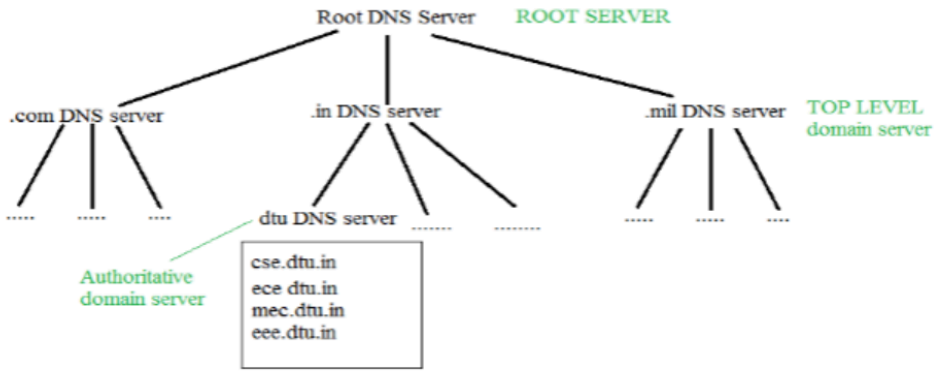
Domain Name System (DNS) is a system used to translate human-readable domain names (e.g., www.example.com) into IP addresses (e.g., 192.0.2.1) that computers use to communicate with each other on the internet. It acts as a distributed database and is an essential service for the smooth operation of the internet. DNS operates at the **Application Layer** of the OSI model.

Why DNS is Needed

Humans find it difficult to remember IP addresses due to their numerical nature. Since IP addresses are also not static (they can change), DNS provides a system that allows the mapping of domain names to IP addresses dynamically. This enables seamless user access to websites by their domain names without worrying about the underlying IP addresses.

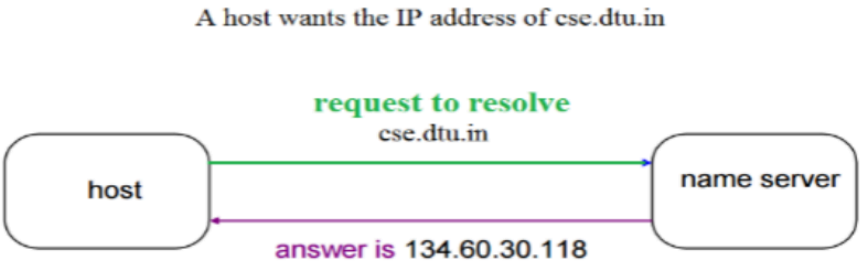
Types of Domains in DNS

1. **Generic Domain**:
    - .com (commercial)
    - .edu (educational)
    - .mil (military)
    - .org (non-profit organizations)
    - .net (similar to commercial)
2. **Country Domain**:
    - .in (India)
    - .us (United States)
    - .uk (United Kingdom)
3. **Inverse Domain**:
    - Inverse mapping is used to determine the domain name from an IP address (reverse DNS lookup).
    - For example, to find the IP address of geeksforgeeks.org, a command like nslookup www.geeksforgeeks.org is used.



DNS Records and Namespace

- **DNS Records** store essential information like domain names, corresponding IP addresses, validity, and Time-to-Live (TTL) values.
- The **Namespace** is the hierarchical structure that defines the set of all possible domain names in the DNS system. A DNS namespace can be organized as a tree, where each level in the hierarchy represents a specific domain segment.



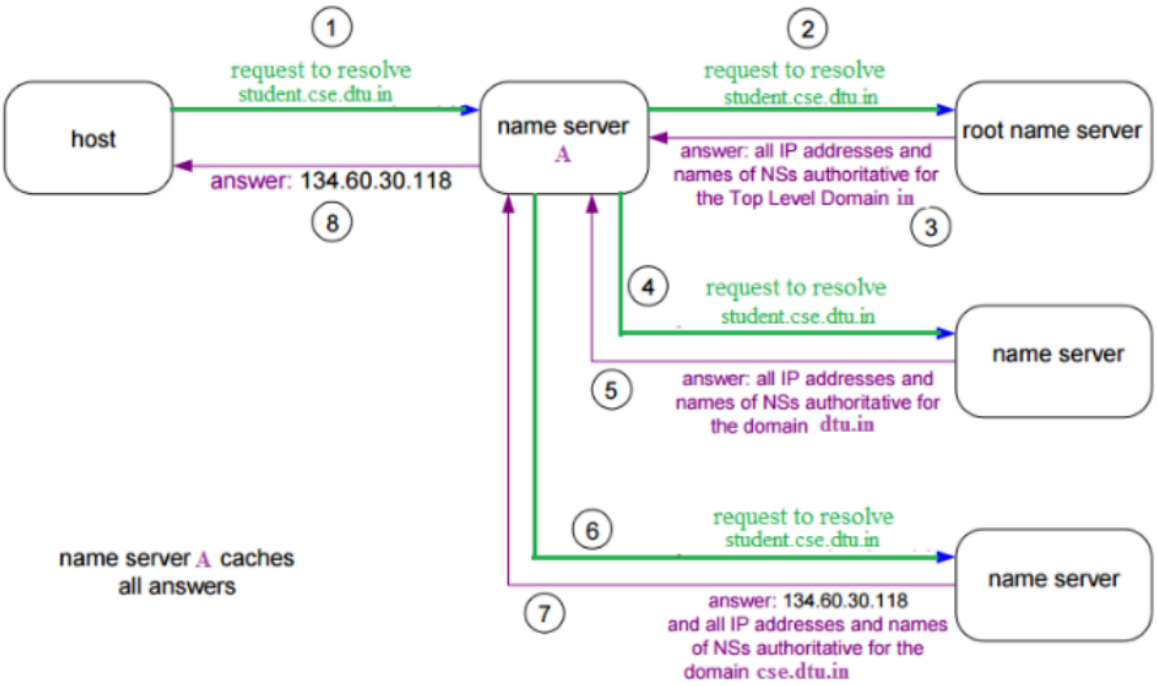
Name-to-Address Resolution

Hierarchical Structure of DNS Servers

1. **Root Name Servers**:
    - These are the top-level DNS servers that manage the global namespace.
    - They provide pointers to the Top-Level Domain (TLD) servers (e.g., .com, .org).
2. **Top-Level Domain (TLD) Servers**:
    - Responsible for the domains like .com, .org, .in, etc.
    - They have information about authoritative name servers for second-level domains.
3. **Authoritative Name Servers**:
    - These are DNS servers that contain the actual mappings for domain names (e.g., www.example.com to 192.0.2.1).
    - These servers can be maintained by the organization or a third-party service provider.
4. **Local Name Servers**:
    - These handle DNS queries for a local network or a particular region, caching results to improve lookup times.

Name to Address Resolution

When a host needs to resolve a domain name (e.g., www.example.com), it sends a query to a DNS server. The server responds with the corresponding IP address. If the server cannot resolve the name, it queries other servers, starting from the root, moving through TLD servers to the authoritative servers until the mapping is found.



Dynamic DNS (DDNS)

The **Dynamic Domain Name System (DDNS)** is a method of automatically updating DNS records in real-time when changes occur, such as when a device's IP address changes. This is especially useful in situations where hosts frequently change their IP addresses (e.g., with home routers).

- **How DDNS works**:
    1. The DNS client sends updates to a primary DNS server when there are changes.
    2. The primary server updates its DNS zone.
    3. Secondary servers (backup DNS servers) are notified and update their records, either actively or passively.
- **Advantages of DDNS**:
    1. Reduces manual updating of static IP addresses.
    2. Provides dynamic address allocation, saving IP address space.
    3. Improves user experience as IP address changes do not affect accessibility.
    4. Seamlessly handles IP changes, maintaining connectivity.
- **Disadvantages of DDNS**:
    1. Less reliable due to the dynamic nature of IP addresses.
    2. Potential security risks as DDNS does not guarantee the authenticity of the device's identity.
- **Uses of DDNS**:
    1. Used by internet-access devices such as home routers.
    2. Used for IP-based devices like DVRs and security appliances.

TELNET :

- Full Form: Terminal Network.
- Purpose: A protocol enabling one computer (local) to connect to another (remote) over TCP/IP for virtual terminal services, defined by ISO.
- Operation:
    - Works on a client-server model.
    - Local computer uses a Telnet client program, and the remote computer uses a Telnet server program.
    - Commands and actions performed on the remote computer are displayed on the local computer.

TELNET Commands:

- Commands are prefixed with IAC (Interpret As Command, code 255), followed by command and option codes.
- Example commands include enabling/disabling options, negotiation, etc.

Modes of Operation:

1. Default Mode:
    - Used if no specific mode is invoked.
    - Client performs echoing.
    - Input is displayed locally but sent to the server only after the entire line is completed.
2. Character Mode:
    - Each character typed is immediately sent to the server.
    - Server echoes characters back for display on the client screen.
3. Line Mode:
    - Line editing (echoing, character erasing) is handled by the client.
    - Entire lines are sent to the server after editing.



Following are some of the important **TELNET commands** :

Character	Decimal	Binary	Meaning
WILL	251	11111011	1. Offering to enable. 2. Accepting a request to enable.
WON'T	252	11111100	1. Rejecting a request to enable. 2. Offering to disable. 3. Accepting a request to disable.
DO	253	11111101	1. Approving a request to enable. 2. Requesting to enable.
DON'T	254	11111110	1. Disapproving a request to enable. 2. Approving an offer to disable. 3. Requesting to disable.

Electronic Mail (Email) Overview:

- Definition: A widely used Internet service that allows users to send messages globally. Messages can include text, images, audio, and video.
- Participants:
    - Sender: Person sending the mail.
    - Recipient: Person receiving the mail.
- Analogy: Functions like a digital version of postal mail.

Components of an Email System:

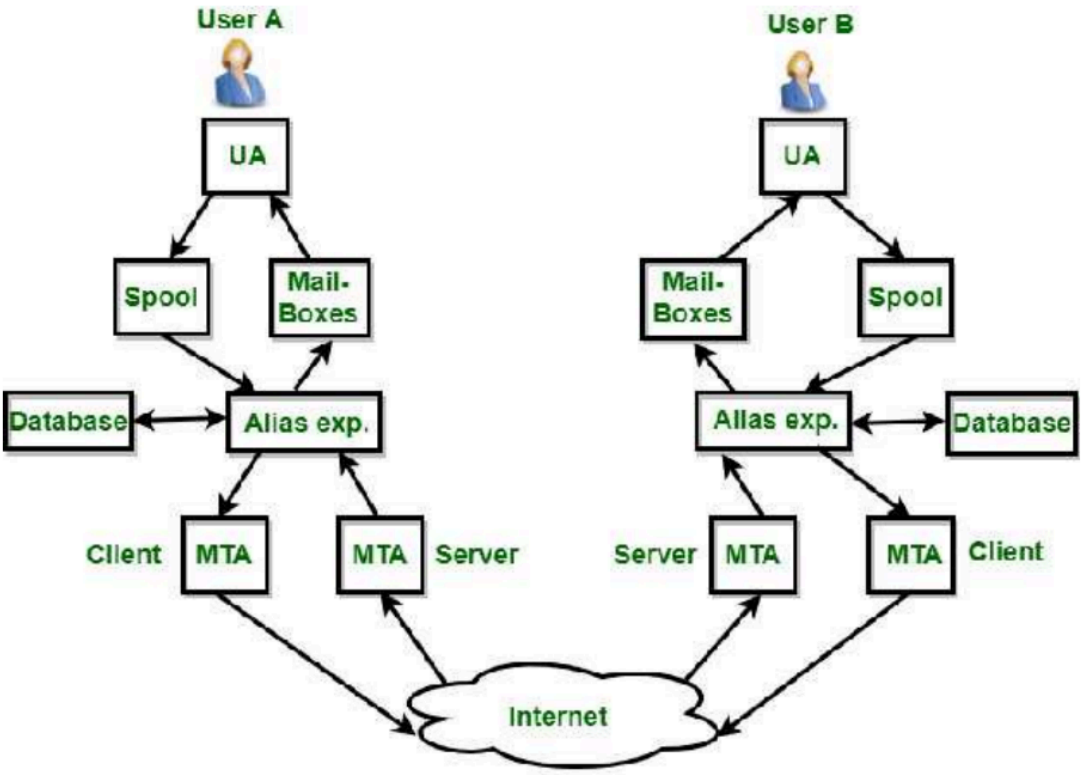
1. User Agent (UA):
    - A program (e.g., mail reader) for composing, sending, receiving, and managing messages.
    - Provides features like spell-check and mail organization.
2. Message Transfer Agent (MTA):
    - Handles mail transfer between systems.
    - Uses SMTP (Simple Mail Transfer Protocol) to deliver emails between MTAs.
3. Mailbox:
    - A file on the local hard drive where received emails are stored.
    - Accessible only to the owner.
4. Spool File:
    - Temporary storage for outgoing emails.
    - MTA extracts emails from this file for delivery.

Email Services:

1. Composition:
    - User agents help compose messages using templates and built-in editors.
2. Reading Messages:
    - Displays a summary of received messages (e.g., sender, subject, status).
3. Replying:
    - Allows responding to senders or all recipients with options to include the original message.
4. Forwarding:
    - Sends messages to third parties, with or without comments.
5. Handling Mailboxes:
    - Supports inbox and outbox for managing received and sent messages, with options for customization.
6. Transfer:
    - Facilitates the delivery process from sender to recipient.
7. Reporting:
    - Confirms delivery status, indicating success, loss, or rejection.
8. Disposition:
    - Lets recipients decide the fate of the email (e.g., save, delete before/after reading).

Key Features:

- Mailing lists enable sending messages to multiple addresses via a single alias.
- Efficient organization and user-friendly interfaces enhance usability.



Advantages of Email:

1. **Cost-effective**: Free email services and minimal charges for online communication.
2. **Accessibility**: Accessible from anywhere with an internet connection.
3. **Convenient**: Allows sending responses at any time.
4. **Speed and Simplicity**: Easy to compose and fast to send.
5. **Mass sending**: Can reach many recipients quickly.

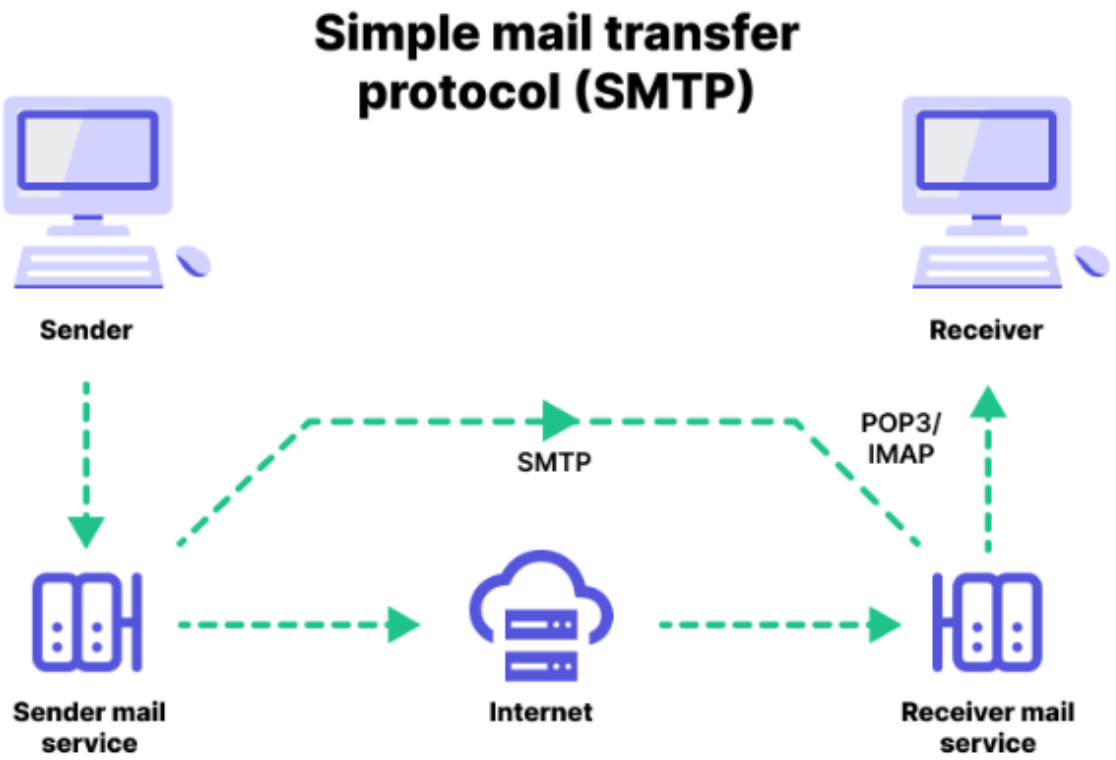
Disadvantages of Email:

1. **Impersonal**: Lacks personal touch compared to face-to-face communication.
2. **Misunderstandings**: Tone and context can be misinterpreted.
3. **Malicious Use**: Can be used for spam or phishing.
4. **Overlong Messages**: Some emails are too lengthy for quick reading.
5. **Insecure**: Sensitive information may be vulnerable to hackers.

Simple Mail Transfer Protocol (SMTP)

The architecture of SMTP (Simple Mail Transfer Protocol) consists of the following components:

1. Mail User Agent (MUA):
    - The email client or application that users interact with to compose and read emails (e.g., Outlook, Gmail).
2. Mail Transfer Agent (MTA):
    - Responsible for routing and transferring emails between servers. The MTA sends, receives, and forwards email messages.
    - Example: Postfix, Sendmail.
3. Mail Delivery Agent (MDA):
    - Delivers the email to the recipient's inbox after it has been processed by the MTA.
    - Example: Procmail, Dovecot.
4. Mail Exchange (MX) Server:
    - The server that handles the receipt of incoming email for a domain, directing it to the appropriate MTA.



SMTP is primarily used for sending email from a client to an email server and between email servers. Here's a basic overview of how it works:

1. Email Composition: The sender writes an email in their email client (e.g., Outlook, Gmail).
2. SMTP Server: The email client sends the composed message to the SMTP server. The server then checks the destination email address.
3. Routing: The SMTP server uses DNS (Domain Name System) to find the recipient's mail server and sends the email to it.
4. Recipient's Mail Server: The recipient's email server (e.g., Gmail, Yahoo) receives the email and stores it in the recipient's inbox.
5. Retrieving Mail: The recipient uses a mail client to retrieve the message via POP3 or IMAP (these protocols are used to fetch emails, unlike SMTP which only sends them).

Multipurpose Internet Mail Extensions (MIME) Protocol :

- Definition: A protocol proposed by Bell Communications in 1991 to extend the capabilities of email by enabling the transfer of non-ASCII data.
- Purpose: Overcomes limitations of SMTP by allowing the exchange of multimedia files (audio, video, images) and applications.

Why MIME is Needed:

1. Limitations of SMTP:
    - Supports only 7-bit ASCII text format.
    - Cannot handle non-ASCII languages (e.g., French, German, Chinese).
    - Lacks support for binary files or multimedia content.
2. Solution:
    - MIME converts non-ASCII data into 7-bit ASCII for transmission and reverts it back on the receiver's side.

Features of MIME:

1. Enables multiple attachments in a single email.
2. Supports unlimited message length.
3. Allows binary attachments (e.g., images, executables, videos) and splits them if necessary.
4. Provides support for varying content types and multipart messages.

MIME Functionality:

1. At the sender's side:
    - Converts non-ASCII data into 7-bit ASCII using MIME protocol.
2. At the receiver's side:
    - Reverts 7-bit ASCII back to the original format for user interpretation.
3. Integration:
    - MIME works with SMTP for transmission and with POP for mailbox organization.

MIME Headers:

Additional headers are added to the email for transformation and identification:

1. MIME-Version: Indicates the MIME protocol version (e.g., 1.0).
2. Content-Type: Specifies the type of content in the email body (e.g., text/plain, image/jpeg).
3. Content-Transfer-Encoding: Describes the encoding method (e.g., 7-bit, 8-bit).
4. Content-ID: Provides a unique identifier for the message.
5. Content-Description: Describes the nature of the content (e.g., image, video, or audio).

MIME Workflow:

- MIME-enhanced messages are encoded into 7-bit ASCII by the sender.
- SMTP delivers the message to the recipient's mailbox.
- POP retrieves and organizes the email for the receiver.
- The MIME header facilitates decoding the message back into its original format for the receiver's user agent.

File Transfer Protocol (FTP) :

FTP (File Transfer Protocol) is a standard network protocol used to transfer files between a client and a server over a network. It allows users to upload, download, and manage files remotely.

Components of FTP:

1. **FTP Client**: The software used by the user to connect to the FTP server, send commands, and transfer files. Examples include FileZilla, WinSCP, or command-line FTP clients.
2. **FTP Server**: The system that stores files and makes them available for transfer. It listens for client requests and handles file uploads, downloads, and management.
3. **Control Connection**: A dedicated channel (typically port 21) used for communication between the client and server, such as sending commands and receiving responses.
4. **Data Connection**: A separate connection used for actual file transfers. This is dynamically established when transferring files and can operate in **active** or **passive** mode.
5. **FTP Commands and Responses**: The client sends commands (like USER, PASS, LIST, RETR, etc.) to the server, which responds with status codes (such as 200 for success or 550 for failure).
6. **Authentication**: Most FTP servers require user authentication (username and password) to grant access, although some servers allow anonymous access.

FTP Architecture:

FTP operates in a **client-server** model:

1. **Client Side**: The user interacts with an FTP client to initiate connections, send commands, and transfer files.
2. **Server Side**: The server hosts the files and listens for requests. It processes commands and responds with data over a dedicated data connection.

   Working:

- The **control connection** is used for sending FTP commands (like login and directory navigation).
- The **data connection** is used to transfer files, such as uploading or downloading data between the client and the server.
- FTP can work in **active mode** (client listens for server connections) or **passive mode** (server listens for client connections), depending on firewall settings.

This simple architecture makes FTP an effective and widely used protocol for file transfer across the internet.

Common FTP Commands:

- USER: Sends user identification.
- PASS: Sends the password.
- CWD: Changes working directory.
- RMD: Removes a directory.
- MKD: Creates a directory.
- PWD: Returns the current directory name.
- RETR: Retrieves a file from the server.
- STOR: Stores a file on the server.
- LIST: Displays files in the directory.
- ABOR: Aborts the current operation.
- QUIT: Terminates the session.

FTP Replies (Examples):

- 200: Command okay.
- 530: Not logged in.
- 331: User name okay, need a password.
- 225: Data connection open; no transfer in progress.
- 221: Control connection closing.
- 503: Bad sequence of commands.

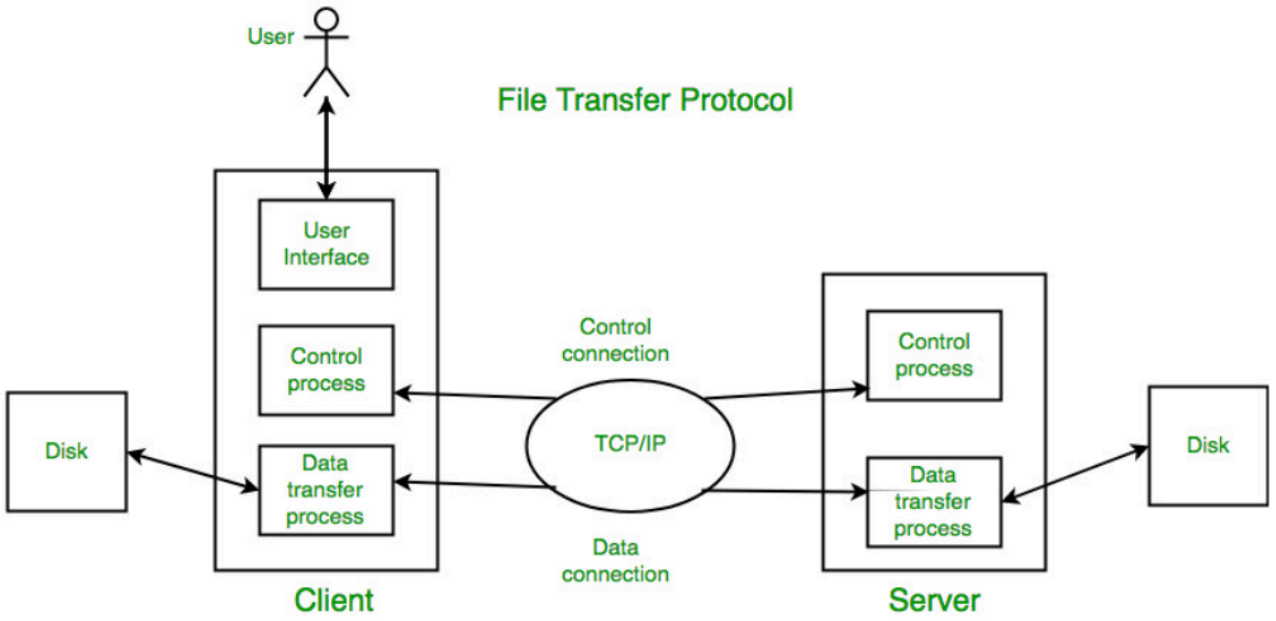
Trivial File Transfer Protocol (TFTP):

- Definition: A simpler file transfer protocol without FTP's advanced features.
- Usage: Suitable for basic file transfers, like boot-time file transfers.
- Characteristics:
    - Uses UDP (port 69) for transport.

- Operates with a simple lock-step protocol where each packet requires acknowledgment.
- Handles errors like packet loss at the server level.

Anonymous FTP:

- Purpose: Allows public access to files without requiring user credentials.
- Login Details:
  - Username: anonymous
  - Password: guest
- Access Restrictions: Limited permissions, such as allowing file downloads but restricting directory navigation.



World Wide Web (WWW)

History:

- Initiated by Tim Berners-Lee in 1989 at CERN for collaborative research.
- Further developed by the World Wide Web Consortium (W3C), directed by Berners-Lee.

System Architecture:

- Web Pages: Linked documents accessible globally.
- Browsers: Software like Chrome, Internet Explorer, and Opera fetch and display pages by interpreting HTML and following hyperlinks.

Working of WWW:

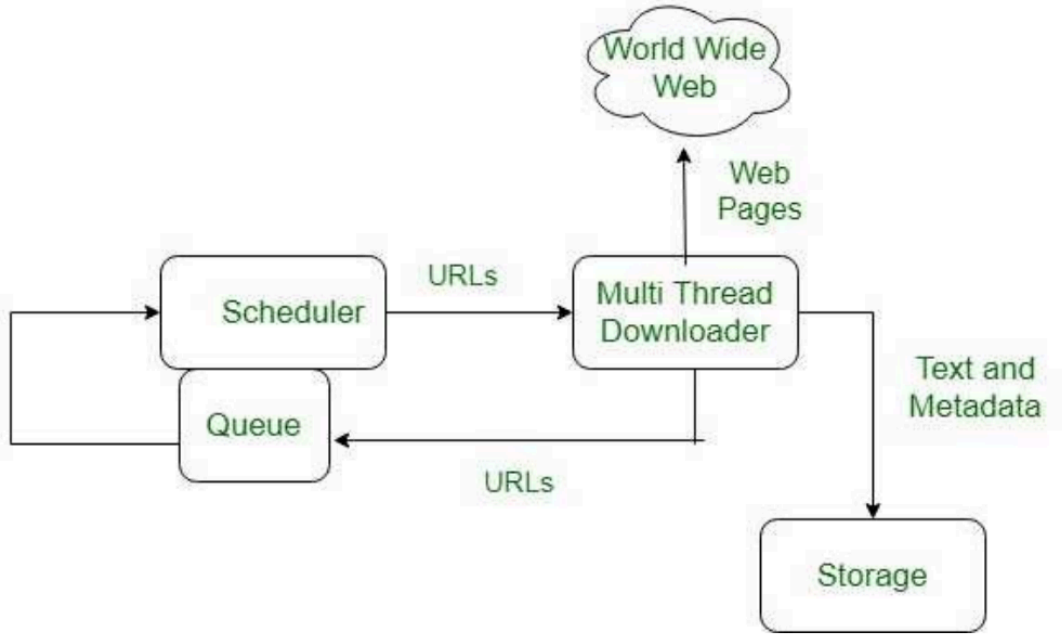
- Based on Web Browsers, HTML, and HTTP.
- Browsers retrieve, display, and navigate hyperlinked resources via a user-friendly interface.

Features of WWW:

1. HyperText Information System
2. Cross-Platform
3. Distributed and Open Standards
4. Dynamic, Interactive, and Evolving
5. Web Browsers as a unified interface

Components of WWW:

1. URL (Uniform Resource Locator): Identifies resources on the web.
2. HTTP (HyperText Transfer Protocol): Governs browser-server communication.
3. HTML (HyperText Markup Language): Defines the content and structure of web pages.



HyperText Transfer Protocol (HTTP) :

HTTP (HyperText Transfer Protocol) is an application layer protocol used for transferring hypertext (coded using HTML) between a web browser (client) and a web server. It is a set of rules for sharing text, images, and multimedia files over the World Wide Web.

How HTTP Works:

- User Action: A user opens a browser and enters a URL (e.g., www.facebook.com).
- DNS Resolution: The Domain Name Server (DNS) translates the URL into an IP address.
- Request: The browser sends a request to the server using the IP address.
- Response: The server processes the request and sends the data (web page) back to the browser.
- Connection Closure: The connection is terminated once the data is exchanged.

Versions of HTTP:

- HTTP 0.9 (1991): Basic request-response protocol for plain text.
- HTTP 1.0 (1996): Introduced metadata and improved functionality (RFC 1945).
- HTTP 1.1 (1997): Enhanced performance, persistent connections, and pipelining (RFC 2616).
- HTTP 2.0 (2015): Optimized for speed with multiplexing and header compression (RFC 7540).
- HTTP 3.0: Based on QUIC, a transport layer protocol developed by Google for faster and secure communication.

Characteristics of HTTP:

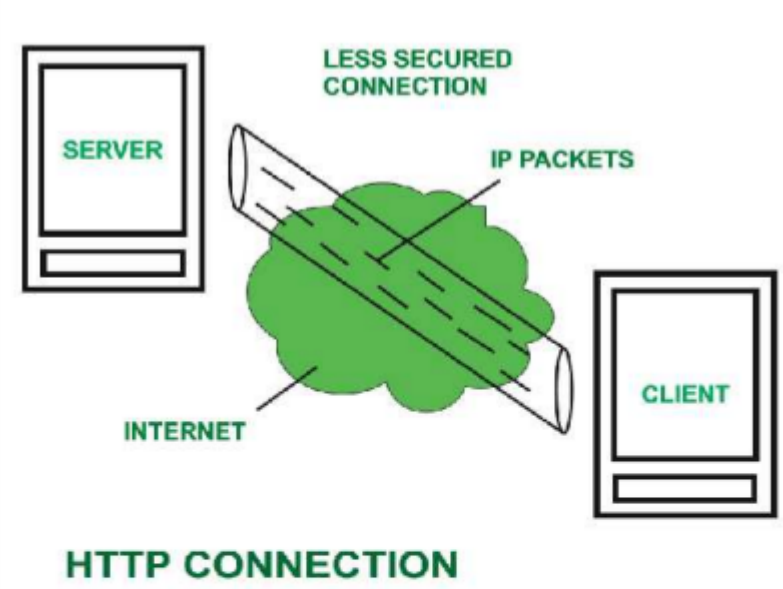
- Request-Response Protocol: Based on client requests and server responses.
- Connectionless: Connections are closed after each request-response cycle.
- Stateless: Neither the client nor server retains memory of past interactions.
- Content Flexibility: Any compatible data type can be exchanged.

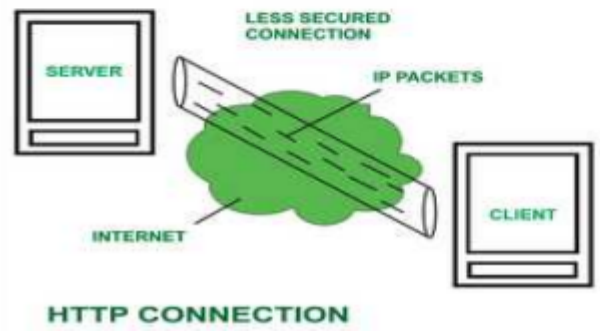
Advantages of HTTP:

- Low memory and CPU usage due to fewer simultaneous connections.
- Reduced latency as handshaking occurs only once per session.
- Supports HTTP pipelining to improve performance.
- Errors can be reported without closing connections.

Disadvantages of HTTP:

- Less Secure: Lacks encryption (e.g., unlike HTTPS that uses TLS).
- High Power Consumption: Requires significant resources for data transfer.
- Not Optimized for Mobile: Can be verbose and inefficient for cellular networks.
- Server Downtime: The server must wait for a client's complete data request, reducing availability for other clients.





Simple Network Management Protocol (SNMP) - :

SNMP is an application layer protocol used to monitor, manage, and configure network devices like routers, switches, and servers. It operates on UDP port 161 (for requests) and UDP port 162 (for traps).

Components of SNMP:

- SNMP Manager:
  - Centralized system, also known as the Network Management Station (NMS).
  - Monitors and controls network devices.
- SNMP Agent:
  - Software installed on managed devices.
  - Collects and provides device data to the SNMP manager.
- Management Information Base (MIB):
  - Hierarchical database containing information about network resources and devices.
  - Organizes data into objects or variables.

SNMP Messages:

- GetRequest: Retrieves specific data from the SNMP agent.
- GetNextRequest: Iteratively discovers all data available on an SNMP agent.
- GetBulkRequest: Fetches large volumes of data in a single request (introduced in SNMPv2c).
- SetRequest: Sets or modifies a variable on the SNMP agent.
- Response: Agent's reply to a manager's request, confirming data retrieval or value update.
- Trap: Unsolicited message sent by an agent to report faults or events.
- InformRequest: Similar to Trap but includes an acknowledgment (introduced in SNMPv2c).

SNMP Security Levels (Used in SNMPv3):

- noAuthNoPriv: No authentication, no encryption. Uses community strings.
- authNoPriv: HMAC with MD5 or SHA for authentication, but no encryption.
- authPriv: HMAC with MD5 or SHA for authentication and DES-56 encryption for privacy.

SNMP Versions:

- SNMPv1:
  - Uses community strings for authentication.
  - Only supports UDP.
- SNMPv2c:
  - Adds GetBulkRequest for bulk data retrieval.
  - Supports UDP by default, configurable for TCP.
- SNMPv3:
  - Most secure version.
  - Uses HMAC with MD5 or SHA for authentication and DES-56 for encryption.
  - Uses TCP for communication.

Bluetooth -Bluetooth is a Wireless Personal Area Network (WPAN) technology used for short-range data exchange. It operates in the 2.4 GHz ISM band and supports a range of up to 10 meters. Bluetooth can connect a maximum of 7 active devices in a network and supports data rates up to 1 Mbps or 3 Mbps (depending on the version).

Firewall

A firewall is a network security system designed to monitor and control incoming and outgoing traffic based on predefined rules. It acts as a barrier between trusted internal networks and untrusted external networks, such as the Internet.

Why Firewalls Are Needed

Before firewalls, Access Control Lists (ACLs) on routers were used for network security. While ACLs could allow or block traffic based on IP addresses and ports, they lacked the ability to:

- Analyze the nature of packets.
- Protect against modern threats.

With the growth of the Internet and its integration into organizations, firewalls became essential to secure internal networks from unauthorized or malicious access.

How Firewalls Work

- Firewalls match network traffic against a predefined rule set.
- Actions applied to traffic depend on matched rules (e.g., allow or block).
- Examples of rules:
    - HR employees cannot access code servers.
    - System administrators can access both HR and technical department data.

Traffic Handling:

- Outgoing traffic is typically allowed but can be restricted for added security.
- Incoming traffic is filtered based on rules for protocols like TCP, UDP, and ICMP.

Default Policy:

- Used when no specific rules apply.
- Best practice: Set the default policy to drop or reject traffic for maximum security.

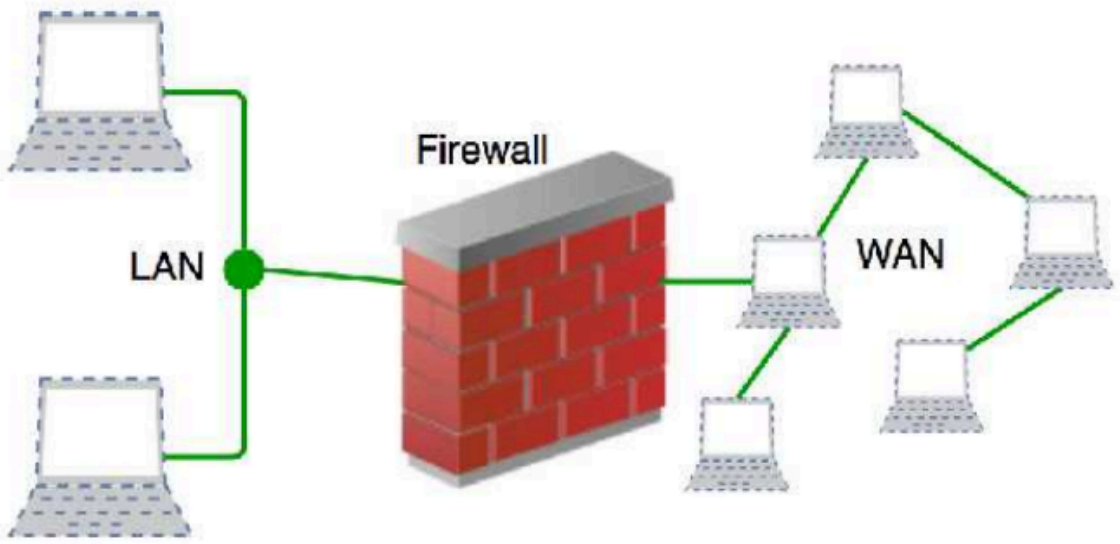
Generations of Firewalls

1. First Generation – Packet Filtering Firewalls:
    - Monitors packets based on source/destination IP, protocols, and ports.
    - Operates at the transport layer of the OSI model.
    - Treats packets independently, with no awareness of traffic state.
    - Maintains a filtering table for traffic control.
    - Example rules:
        - Block packets from 192.168.21.0.
        - Allow well-known services to 192.168.21.0.
2. Second Generation – Stateful Inspection Firewalls:
    - Tracks the state of connections (e.g., TCP streams).
    - Decisions are based on rules and the state table (packet history).
    - More efficient than packet filtering firewalls.

3. Third Generation – Application Layer Firewalls:
    - Inspects and filters packets at any OSI layer, including the application layer.
    - Can block specific content and detect misuse of applications/protocols.
    - Acts as a proxy server, preventing direct connections.
    - Can function as a Network Address Translator (NAT).
4. Next-Generation Firewalls (NGFWs):
    - Provide Deep Packet Inspection, application monitoring, and SSL/SSH inspection.
    - Address advanced threats like malware and application-layer attacks.

Types of Firewalls

1. Host-Based Firewalls:
    - Installed on individual devices (e.g., laptops, servers).
    - Provides protection within trusted networks.
    - Comes as part of operating systems or standalone software.
2. Network-Based Firewalls:
    - Operates at the network level, filtering traffic for the entire network.
    - Typically hardware devices with proprietary software.
    - May have multiple network interface cards (NICs).



Do Port Addresses Need to Be Unique?

- **Locally Unique**: On a single device, port numbers must be unique to avoid conflicts between applications.
- **Globally Not Unique**: Across the internet, the combination of IP address and port number ensures uniqueness.

Why Are Port Numbers Shorter Than IP Addresses?

1. **Purpose**: Port numbers identify services on a device, while IP addresses identify devices on a network.
2. **Size**: Port numbers are 16 bits (0–65535), enough to manage services efficiently, while IP addresses are longer (32 bits for IPv4, 128 bits for IPv6) to identify billions of devices.
3. **Efficiency**: Smaller port numbers reduce processing overhead and improve performance.

Why Do We Need DNS Instead of Directly Using IP Addresses?

1. **Ease of Use**: DNS translates easy-to-remember domain names (e.g., google.com) into IP addresses.
2. **Dynamic IPs**: DNS manages changing IP addresses without user disruption.
3. **Scalability**: Supports load balancing and directs users to the nearest servers for better performance.
4. **Management**: Simplifies website maintenance by allowing IP changes without affecting users.
5. **Shared IPs**: Enables multiple domains to share one IP address.
6. **Redundancy**: Ensures high availability through distributed servers.

DNS makes the internet user-friendly, flexible, and efficient.

Why FTP Does Not Have a Message Format

FTP (File Transfer Protocol) focuses on **file transfer**, not message formatting. Unlike email protocols like SMTP, FTP does not need a predefined message format because:

1. **Command-Response Structure**:
    - FTP uses simple, predefined commands (e.g., USER, RETR, STOR) and responses (e.g., 200 OK, 550 Error) to control file transfers.
2. **Data Transfer Focus**:
    - FTP separates control (commands) and data (files). Files are transferred as raw data without needing a specific message format.
3. **No Content Interpretation**:
    - FTP does not interpret or structure the content; it simply moves files between client and server.

Thus, FTP prioritizes efficient file transfer rather than formatting messages.